Physics Group Project Report

An Explanatory Guide and Step-by-Step Derivation of
"The Pendulum — Rich Physics from a Simple System"

Based on the article by R. A. Nelson & M. G. Olsson,
American Journal of Physics **54**(2), 112–121 (1986)

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Abstract

This report unpacks the foundational paper "The Pendulum — Rich Physics from a Simple System" by Nelson and Olsson. The original article shows how the acceleration of gravity g can be measured to one part in 10^4 with a plane pendulum, provided a long list of non-ideal corrections is applied. The published derivations frequently compress several algebraic steps into a single line, so here we fill those steps in explicitly, explain the physics behind each correction, and organize the results so an introductory student can follow them. We then describe our own small-scale reproduction of the experiment in Daegu, South Korea. Using a 0.788 m thread pendulum and a smartphone stopwatch, we measured $g = 10.08 \pm 0.19$ m/s^2, about 2.9 % above the accepted local value of 9.797 m/s^2. More importantly, we directly observed the finite-amplitude effect predicted by the paper: the period grows with release angle, and applying the finite-amplitude correction removes the angle dependence from our extracted value of g.

Contents

1 Introduction and Contextual Background

The plane pendulum is one of the oldest precision instruments in physics. Its approximate isochronism — first noted by Galileo — made it the basis of accurate mechanical clocks, and later work using pendulums provided early evidence that inertial and gravitational mass are equivalent. For a long period the pendulum was also the most practical laboratory method for measuring local gravity. Nelson and Olsson revisit this "simple" system and show that, once high accuracy is demanded, the idealized point-mass description is no longer adequate and a surprisingly rich collection of physical effects must be accounted for.

1.1 Historical and Pedagogical Significance

The central thesis of the paper is that determining g to four significant figures with an ordinary laboratory pendulum is possible, but only if one systematically computes and applies corrections for finite swing amplitude, the finite size and mass distribution of the bob and suspension, air buoyancy, damping, added mass, and the elasticity of the wire and support. The intended audience is upper-level undergraduate and introductory laboratory students: the paper is valuable precisely because it does not stop at simple harmonic motion (as most textbooks do) but instead treats the pendulum as a real, slightly messy object and shows how each imperfection enters the period. This is the pedagogical move we found most useful, and it is the spirit in which we carried out our own experiment.

1.2 Physical Setup and System Parameters

Table 1 lists the symbols used throughout. Where it matters we distinguish the apparatus in the original paper (a 3 m pendulum read with a cathetometer) from our own apparatus (a 0.79 m thread pendulum read with a stopwatch).

2 Step-by-Step Mathematical Derivations

2.1 The Starting Framework

A point mass on a massless, inextensible cord of length l swinging in a vacuum obeys the torque equation

$$\ddot{\theta} + \frac{g}{l}\sin\theta = 0. \tag{1}$$

For infinitesimal displacements we expand $\sin\theta \approx \theta$, and Eq. (1) reduces to the simple-harmonic form $\ddot{\theta} + \omega_0^2\theta = 0$ with $\omega_0^2 = g/l$. The solution oscillates at the natural frequency ω_0, so the

Table 1: Key symbols, constants, and variables.

Symbol	Physical meaning	SI unit
m	Mass of the bob	kg
a	Radius of the spherical bob	m
l (L_{eff})	Effective length, pivot to centre of bob	m
θ	Angular displacement	rad
θ_0	Maximum (release) angular displacement	rad
T	Measured period of oscillation	s
T_0	Idealized (small-angle) period	s
ω_0	Natural angular frequency	rad/s
I	Moment of inertia about the pivot	$\text{kg}\,\text{m}^2$
ρ	Density of air	kg/m^3
g	Acceleration due to gravity	m/s^2

idealized period is

$$T_0 = 2\pi\sqrt{\frac{l}{g}} = \frac{2\pi}{\omega_0}. \tag{2}$$

Rearranging Eq. (2) gives the working equation that converts a measured length and period into gravity:

$$g = \frac{4\pi^2 l}{T_0^2}. \tag{3}$$

2.2 Filling in the Missing Step: the Correction Framework

A real pendulum never has the ideal period. The paper writes the measured period as the ideal period plus a cumulative offset,

$$T = T_0 + \Delta T = T_0\left(1 + \frac{\Delta T}{T_0}\right). \tag{4}$$

To recover the ideal T_0 from the measured T we divide and expand. Because every individual correction is tiny ($\Delta T/T_0 \ll 1$), a first-order binomial expansion $(1+x)^{-1} \approx 1 - x$ is justified:

$$T_0 = \frac{T}{1 + \Delta T/T_0} \approx T\left(1 - \frac{\Delta T}{T_0}\right). \tag{5}$$

The strategy of the paper is therefore: measure T, compute every contribution to $\Delta T/T_0$ separately, add them (they are small enough to add linearly), and feed the corrected T_0 into Eq. (3).

3 Physical Corrections and Limiting Cases

This section reconstructs the dominant corrections of the paper, evaluated for the *original* apparatus ($a = 3.05\,\text{cm}$, $l = 300.44\,\text{cm}$, $\theta_0 = 3.0^\circ$). We later show in Section 4 that, at our much coarser precision, only the first of these is large enough to matter.

3.1 Finite-Amplitude Correction

The linear approximation $\sin\theta \approx \theta$ breaks down for finite swings. An exact treatment of Eq. (1) via energy conservation gives the period as a complete elliptic integral of the first kind, $K(k)$, with $k = \sin(\theta_0/2)$:

$$T = \frac{4}{\omega_0} K\left(\sin \tfrac{\theta_0}{2}\right). \tag{6}$$

Expanding K as a power series and integrating term by term yields

$$\frac{\Delta T}{T_0} = \frac{1}{16}\theta_0^2 + \frac{11}{3072}\theta_0^4 + \cdots \tag{7}$$

For $\theta_0 = 3.0°$ the leading term gives $\Delta T \approx +596\,\mu s$: larger swings genuinely take longer. This is the single most important correction and the one we were able to observe directly in our own data.

3.2 Mass-Distribution Corrections

A real bob has finite size, so the system is a *physical* pendulum with period

$$T = 2\pi\sqrt{\frac{I}{Mgh}}, \tag{8}$$

where I is the moment of inertia about the pivot, M the total mass, and h the pivot-to-centre-of-mass distance. For a uniform sphere of radius a on a massless cord, the parallel-axis theorem gives $I = ml^2\left[1 + \frac{2}{5}(a/l)^2\right]$ with $M = m$, $h = l$, so

$$\frac{\Delta T}{T_0} = \frac{1}{5}\left(\frac{a}{l}\right)^2. \tag{9}$$

For the paper's apparatus this is $+72\,\mu s$. The suspension ring, attachment cap, cap screw, and the distributed mass of the wire each add further small terms; the wire mass alone shortens the period by $\Delta T/T_0 = -\frac{1}{12}(m_w/m)$, about $-463\,\mu s$ in the original experiment.

3.3 Air Corrections

The surrounding air contributes in three ways. *Buoyancy* reduces the bob's apparent weight (Archimedes), which weakens the effective restoring "gravity" and lengthens the period:

$$\frac{\Delta T}{T_0} = \frac{1}{2}\frac{m_a}{m}, \tag{10}$$

giving $+292\,\mu s$ for the paper. *Damping* (a combination of linear and quadratic air drag) slightly increases the period, and *added mass* — the air dragged along with the bob — increases the system's effective inertia, contributing about $+346\,\mu s$.

3.4 Elastic Corrections

Finally, neither the wire nor the support is perfectly rigid. Static and dynamic stretching of the wire and small lateral motion of the mounting point each shift the period, but for the paper's stiff steel wire and massive support these terms are at the microsecond level or below.

3.5 Summary of the Paper's Corrections

Table 2 collects the corrections for the original apparatus. Their sum, $\Delta T = +472\,\mu s$, converts the naive measurement into $T_0 = 3.478\,33\,s$ and hence $g = 9.8034(14)\,m/s^2$, in agreement with the independently known local value — the headline success of the paper.

Table 2: Dominant corrections for the original Nelson–Olsson apparatus.

Effect	ΔT (µs)
Finite amplitude	$+596$
Finite radius of bob	$+72$
Mass of ring	-282
Mass of cap	-116
Mass of cap screw	$+95$
Mass of wire	-463
Buoyancy	$+292$
Added mass	$+346$
Damping (net)	~ -44
Elastic / support	$\sim +4$
Total	$\mathbf{+472}$

4 Experimental Reproduction and Results

To put the paper's framework into practice we built and timed a simple pendulum and extracted g from our own data.

4.1 Apparatus and Setup

The bob was a steel ball of mass $m = 200.0(1)$ g and radius $a = 1.35$ cm, suspended on a thin thread. The effective length, measured from the pivot to the centre of the ball after the setup was finalized, was $L_{\text{eff}} = 78.8$ cm. The experiment was carried out in Daegu, South Korea, where the accepted local gravity is $9.797\,\text{m/s}^2$.

4.2 Engineering Iterations

Our raw setup did not behave ideally, and dealing with this is itself an illustration of the paper's theme that a real pendulum departs from the ideal. Three problems and fixes are worth recording. (i) A loosely tied knot let the bob pivot about a second point, so the system briefly behaved as a *double pendulum*; inserting a small wood piece at the knot fixed the pivot and stopped this motion. (ii) The thread swung slightly forward and back rather than in a single plane; closing the gap between the support and the thread constrained it. (iii) Releasing the bob by hand introduced sideways motion, so we released it against a foam board for a consistent, push-free start. Each fix made the timing more repeatable, and — importantly — the knot/support changes are what set the final effective length.

4.3 Procedure and Timing Data

For each release angle we timed $N = 20$ complete oscillations and repeated for six trials. We used angles of 5°, 10°, 15°, and 20°, with a single 30° trial. Table 3 reports the mean time, sample standard deviation, the mean period $\bar{T} = \bar{t}/20$, the finite-amplitude factor from Eq. (7), and the corrected small-angle period $T_0 = \bar{T}/(1 + \theta_0^2/16 + 11\theta_0^4/3072)$.

The 30° point has only one trial, so we exclude it from the final analysis and use it only as a qualitative consistency check.

Table 3: Timing data and finite-amplitude correction ($N = 20$ oscillations).

θ_0	$\bar{t}_{20}$ (s)	s (s)	$\bar{T}$ (s)	amplitude factor	T_0 (s)
5°	35.150	0.079	1.7575	1.000476	1.7567
10°	35.128	0.069	1.7564	1.001907	1.7531
15°	35.343	0.137	1.7672	1.004301	1.7596
20°	35.435	0.044	1.7718	1.007669	1.7583
30°	36.770	—	1.8385	1.017404	1.8071

4.4 Extracting g

Applying $g = 4\pi^2 L_{\text{eff}}/T^2$ to both the *raw* mean period and the *corrected* period gives Table 4. The contrast between the two columns is the key result, shown in Fig. 1.

Table 4: Gravity extracted from the raw and corrected periods.

θ_0	g_{raw} (m/s^2)	$g_{\text{corrected}}$ (m/s^2)
5°	10.072	10.081
10°	10.084	10.123
15°	9.962	10.048
20°	9.910	10.063
mean	10.007	10.079
spread (SD)	0.085	0.032

4.5 Final Value and Error Analysis

Averaging the four corrected angles gives

$$g = 10.08 \pm 0.19 \text{ m/s}^2, \tag{11}$$

about +2.87 % relative to the accepted 9.797 m/s^2, which lies roughly 1.5 standard deviations below our mean. The uncertainty is dominated by the effective length: propagating

$$\left(\frac{\sigma_g}{g}\right)^2 = \left(\frac{\sigma_L}{L_{\text{eff}}}\right)^2 + \left(\frac{2\sigma_T}{T_0}\right)^2, \tag{12}$$

the timing term is negligible (our corrected g values agree across angles to ±0.03), whereas a realistic length uncertainty of $\sigma_L \approx 1.5$ cm — arising from the difficulty of locating the true pivot after the knot and support were adjusted — accounts for essentially the entire error bar.

4.6 Which Corrections Matter at Our Scale

A useful exercise is to ask which of the paper's corrections are even relevant for our crude apparatus. The finite-amplitude term is large: between 5° and 20° it shifts the period by about 0.7 %, which is exactly the trend removed in Fig. 1. By contrast, the finite-bob-radius term of Eq. (9) evaluates to $\frac{1}{5}(1.35/78.8)^2 \approx 6 \times 10^{-5}$, i.e. a 0.006 % effect on the period, and the wire-mass and buoyancy terms are similarly small. We therefore applied only the finite-amplitude correction and neglected the rest — a deliberate choice that reflects the precision of a stopwatch-and-thread experiment, rather than an oversight.

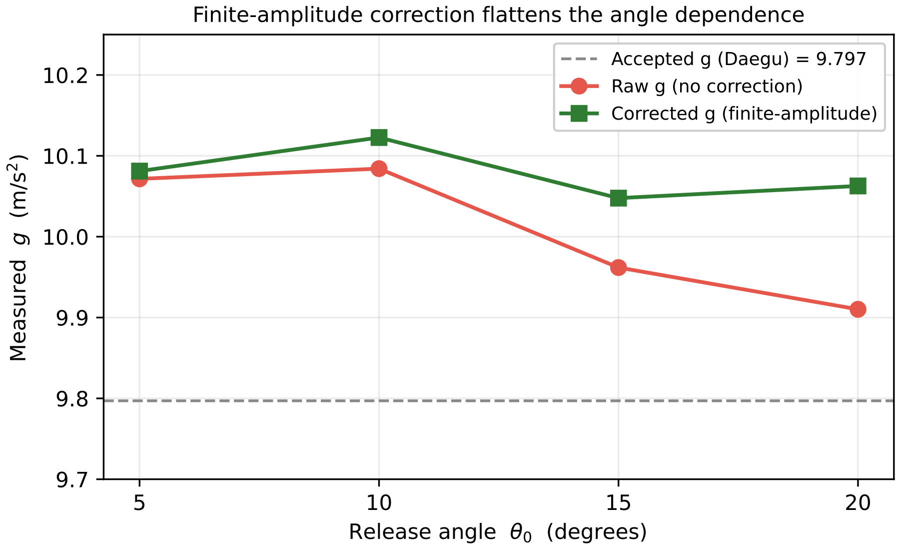

Figure 1: Gravity extracted at each release angle. The raw values (red) fall as the angle increases, because larger swings have longer periods. Applying the finite-amplitude correction (green) removes this trend and leaves a nearly flat value of g. Both lie about 3 % above the accepted local value (dashed), reflecting the dominant uncertainty in the effective length.

5 Critical Analysis and Commentary

The idealized formulae rest on several assumptions: negligible air resistance and buoyancy, a perfectly rigid and inextensible suspension, a point-mass bob, and a fixed pivot. The paper's real contribution is to show that these assumptions are not merely approximations to be waved away but each correspond to a measurable, calculable shift in the period. At small angles and modest precision the assumptions hold well, but as accuracy demands grow, the non-ideal effects become significant and the analysis must become correspondingly more realistic.

A second implicit assumption is that the measurements themselves are exact. In practice the length, period, mass, and damping all carry uncertainty, and our own experiment makes the consequences concrete: our result is limited not by timing but by how well we could define the effective length. The paper is somewhat light on this practical side — it focuses on the theory of the corrections rather than on the day-to-day difficulty of building a clean pendulum — and our experience suggests that connecting the elegant corrections to the messiness of real data collection deserves more emphasis.

The strength of the paper is pedagogical: it demonstrates how a deceptively simple system can display rich behaviour once examined carefully, and how a basic model can be refined step by step into a precision instrument. The same logic extends well beyond pendulums — to seismometers, mechanical clocks, earthquake engineering, and any oscillatory system where damping and nonlinearity matter.

6 Conclusion

We reconstructed the central derivations of Nelson and Olsson, supplying the intermediate algebra the paper compresses, and explained the physical origin of each correction to the ideal pendulum period. We then reproduced the experiment at introductory scale. Our measured value, $g = 10.08 \pm 0.19$ m/s^2, is about 2.9 % high, with the discrepancy traceable to the uncertainty in the effective length rather than to the timing. Most satisfying, our multi-angle data reproduced the paper's central qualitative claim: the period lengthens with amplitude, and the finite-amplitude correction — the only one large enough to matter at our precision — flattens the angle dependence in the extracted g. Working through both the theory and a hands-on measurement made clear why even a "simple" pendulum repays careful analysis, and how unpacking each approximation connects the mathematics to the behaviour of a real physical system.

Code and Data Availability

The interactive simulation, raw timing data, this report, and its LaTeX source are available at:

`https://github.com/Iveel17/physics_project`

A live interactive demonstration is hosted at `https://iveel17.github.io/physics_project/`.

References

[1] R. A. Nelson and M. G. Olsson, "The pendulum — Rich physics from a simple system," *American Journal of Physics* **54**(2), 112–121 (1986).

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[3] J. B. Marion and S. T. Thornton, *Classical Dynamics of Particles and Systems*, 5th ed. (Brooks/Cole, 2003).